

LAB 07

PH142 Summer 2026

Announcements

- 7/28: Quiz16, Lab07
- 7/29: Quiz17
- 7/30: Midterm 2
- 7/31: Quiz18, Data Project Part 2*

*Your group must meet with your assigned GSI before the due date



Lab07 Lecture Review

Definitions

Parameter (μ): A fixed number that describes the population (usually unknown)

Statistic ($\bar{x}$): A number calculated from the sample; used to estimate the parameter

- Different samples give different $\bar{x}$, the statistic varies and is a random variable

Lab07 Lecture Review

Definitions

Sampling Distribution: The distribution of sample means, $\bar{x}$, from all possible samples of size n .

- Centered at the true mean, μ
- Has a standard deviation of: $\frac{\sigma}{\sqrt{n}}$

Lab07 Lecture Review

Central Limit Theorem (CLT)

Definition: When n is large, the sampling distribution of the mean $\bar{x}$ is approximately Normal. The shape of the original population distribution does not matter!

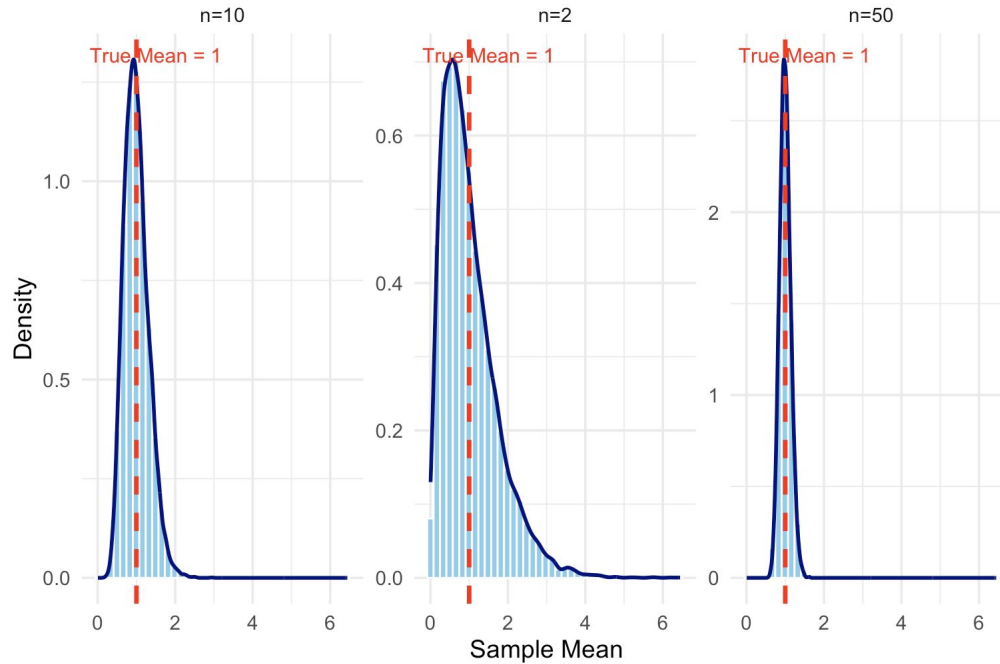
As n increases...

- The variability of $\bar{x}$ decreases
- The distribution of $\bar{x}$ becomes more Normal

Lab07 Lecture Review

The Central Limit Theorem in Action

The sampling distribution becomes more Normal as n increases, even from a skewed



Lab07 Lecture Review

Confidence Intervals for a Mean

When we use the **sample mean** ($\bar{x}$) to estimate the true **population mean** (μ), each sample will be slightly different.

A confidence interval provides a range of values that likely contain μ . The CI reflects our estimate with some uncertainty/variability, and is given by:

$$\text{CI} = \text{Point estimate} \pm \text{Margin of error}$$

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Confidence Intervals for a Mean

CI Formula: $\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$

$\bar{x}$ = sample mean

z^* = critical value (1.96 for 95% CI)

$\frac{\sigma}{\sqrt{n}}$ = standard error of the mean



LAB 07 Walkthrough

Lab Submission

- Follow the directions on the LAB07 file
- Submit using the **Terminal Tab** (next to the console in the bottom left pane)
 - Copy and paste the given line into the terminal
 - Follow prompts (NOTE: the terminal will **not** show your password being typed out!)
- **CHECK IN GRADESCOPE THAT ALL YOUR TESTS PASSED**